

Competency

- To construct square, rectangle, rhombus, parallelogram, quadrilateral and trapezium and analyze them.

Learning Outcomes

- To construct square, rectangle, rhombus, parallelogram, quadrilateral and trapezium.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Application (A)	- To sketch the rough figure and construct square, rectangle, rhombus, parallelogram, quadrilateral and trapezium under the given information

Required Teaching Materials/ Resources

Geometric instruments, geo-board, pencils, marker, ICT tool etc.

Pre-knowledge: properties of square, rectangle, rhombus, parallelogram, quadrilateral and trapezium.

Teaching Activities

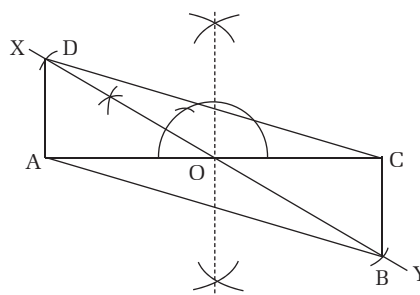
1. Discuss about the procedures of constructions of square and give the different side / diagonal lengths of squares to construct in groups.
2. Discuss about the procedures of constructions of rectangle and rhombus and give the different information to construct in groups.
3. Explain the constructions procedures of quadrilateral and give the different information to construct in groups.
4. Explain the constructions procedures of trapezium and give the different information to construct in groups.

Solution of selected problems from Vedanta Excel in Mathematics

1. Construct a parallelogram ABCD in which diagonals $AC = 4.6$ cm, $BD = 5.8$ cm and they bisect each other making an angle of 30° .

Steps of construction

- (i) Draw a diagonal $AC = 4.6$ cm,
 - (ii) Draw the perpendicular bisector of AC and mark its mid-point O.
 - (iii) At O, construct $\angle AOX = 30^\circ$ and produce XO to Y.
 - (iv) Here, O is also the mid-point of BD. With centre at O and radius $OB = OD = 2.9$ cm ($\frac{1}{2}$ of BD) cut OY at B and OX at D.
 - v) Join A, D; B, C; A, B and C, D.
- Thus, ABCD is the required parallelogram.

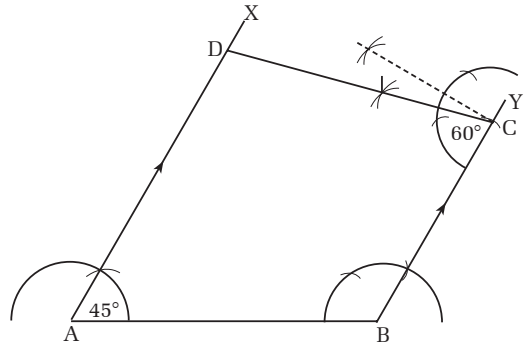


2. Construct a trapezium ABCD in which $AB = 5.5$ cm, $BC = 4.5$ cm, $\angle DAB = 45^\circ$, $\angle BCD = 60^\circ$ and $AD \parallel BC$.

Steps of construction

- Draw a line segment $AB = 5.5$ cm.
- Construct $\angle BAX = 45^\circ$ at A.
- D lies on AX and $AD \parallel BC$.
So, at B, construct
 $\angle ABY = 180^\circ - 45^\circ = 135^\circ$
- With the centre at B and radius 4.5 cm draw an arc to cut BY at C.
- At C, construct $\angle BCD = 60^\circ$. The arm CD intersect AX at D.

Thus, ABCD is the required trapezium.

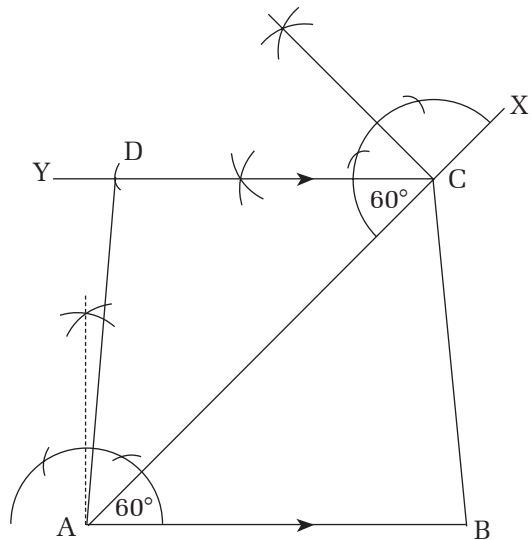


3. Construct a trapezium ABCD in which $AB = 4.8$ cm, diagonal $AC = 5.9$ cm, $\angle BAC = 60^\circ$, $CD = 5$ cm and $AB \parallel DC$.

Steps of construction

- Draw a line segment $AB = 4.8$ cm.
- At A, construct $\angle BAX = 60^\circ$.
- With centre at A and radius 5.9 cm, draw an arc to cut AX at C.
- Join B and C.
- As $AB \parallel DC$, alternate angles BAC and ACD are equal. So, construct $\angle ACY = 60^\circ$ at C.
- With centre at C and radius 5 cm, draw an arc to cut CY at D.
- Join D and A.

Thus, ABCD is the required trapezium.



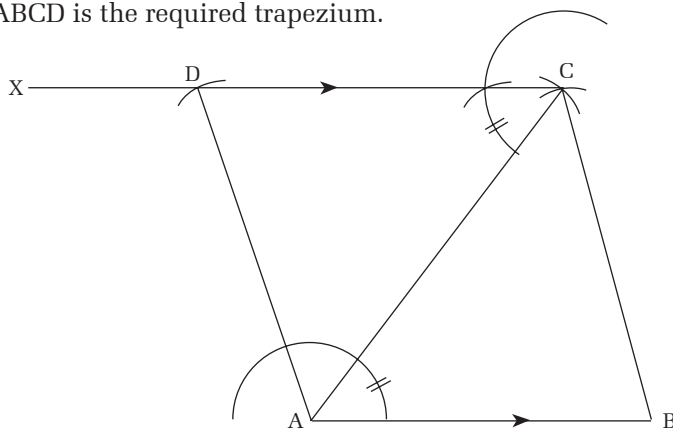
4. Construct a trapezium ABCD in which $AB = 4.4$ cm, diagonal $AC = 6.8$ cm, $AD = BC = 5.2$ cm and $AB \parallel DC$.

Steps of construction

- Draw a line segment $AB = 4.4$ cm.
- With centre at A and radius 6.8 cm, draw an arc.
- With centre at B and radius 5.2 cm, draw another arc to intersect the previous arc at C.

- (iv) Join A, C and B, C.
- (v) At C, construct $\angle ACX = \angle BAC$.
- (vi) With centre at A and radius 5.2 cm, draw an arc to cut CX at D.
- (vii) Join A and D.

Thus, ABCD is the required trapezium.



5. **Construct a quadrilateral ABCD in which $AB = 5$ cm, $BC = 5.6$ cm, $CD = 4.5$ cm, $AD = 5.4$ cm and the diagonal $BD = 6.5$ cm.**

Steps of construction

- (i) Draw $AB = 5$ cm.
 - (ii) From A, draw an arc with radius $AD = 5.4$ cm and from B draw another arc with radius $BD = 6.5$ cm. These two arcs intersect each other at D.
 - (iii) From B, draw an arc with radius $BC = 5.6$ cm and from D, draw another arc with radius $DC = 4.5$ cm. These two arcs intersect to each other at C. Join A, D; B, C and D, A.
- Thus, ABCD is the required quadrilateral.

